
Linear Mode Connectivity and Model Merging: Permutation Alignment, REPAIR, and Multi-Model Fusion across Architectures

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Abstract

Independently trained networks that reach the same low loss are typically separated by a high *loss barrier* along their linear interpolation, despite living in a single connected loss basin once the permutation symmetry of hidden units is accounted for. We present a unified empirical study of linear mode connectivity (LMC) and weight-space model merging across three architecture/dataset pairs deliberately chosen to differ from the course notebooks: an MLP on KMNIST, a custom BatchNorm ConvNet on SVHN, and MobileNetV3-small on EuroSAT. We (i) reduce the interpolation barrier by neuron alignment (weight matching, activation matching, Procrustes), (ii) explain and correct the *variance collapse* of interpolated activations with REPAIR, (iii) visualise the 2D loss landscape before and after alignment, (iv) scale merging to $n \in \{3, 5, 8\}$ models with anchor and iterative alignment, and (v) study task arithmetic and TIES-merging. Alignment lowers the MLP barrier from 0.91 to $\approx 10^{-3}$, iterative alignment keeps the barrier near zero up to eight models, and scaled task vectors recover 87% joint accuracy where naive addition collapses to chance.

1. Introduction

Stochastic gradient descent finds many distinct minima of the training loss that all generalise well. A natural question is how these minima relate geometrically. *Linear mode connectivity* (LMC) asks whether two solutions θ_A, θ_B can be joined by a straight line of low loss. Empirically the interpolated model $\theta(\alpha) = \alpha\theta_A + (1 - \alpha)\theta_B$ usually incurs a large *loss barrier* at $\alpha = 0.5$, suggesting the two minima sit in different valleys. Entezari et al. (2022) conjectured, and Ainsworth et al. (2023) demonstrated, that much of this barrier is an artefact of the *permutation symmetry* of neurons: by relabelling the hidden units of one network to match the other, the two solutions fall into a shared basin and the barrier nearly vanishes.

This geometric fact underpins *model merging*: combining several independently trained networks directly in weight space, with no access to the original data and no extra inference cost. Merging is the basis of model soups (Wortsman et al., 2022), federated and distributed averaging, and task arithmetic for model editing (Ilharco et al., 2023). In this project we build a single, self-contained pipeline that connects these threads. Our goal is not a new algorithm but a careful, reproducible characterisation of *when* weight-space merging works, *why* it fails, and *how* alignment and activation renormalisation fix it, across architectures that stress different parts of the theory (no normalisation, BatchNorm, and a modern mobile backbone). All code is available at <https://github.com/carlosgr1/DeepLearning-AppliedAI>.

2. Related work

Frankle et al. (2020) introduced LMC in the context of the lottery ticket hypothesis, and Garipov et al. (2018) showed that non-linear low-loss paths almost always exist. Entezari et al. (2022) traced the linear barrier to permutation invariance, and Git Re-Basin (Ainsworth et al., 2023) gave practical weight- and activation-matching algorithms that solve a linear assignment problem (LAP) per layer. REPAIR (Jordan et al., 2023) identified that interpolation also

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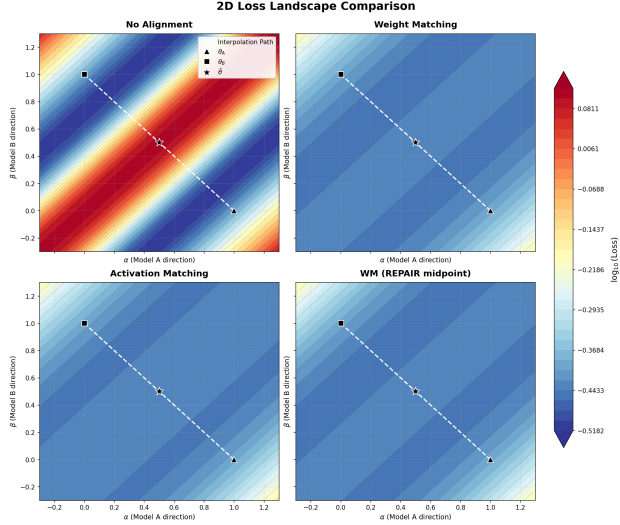


Figure 1. 2D loss landscape (MLP-3 / KMNIST) spanned by the two endpoints θ_A, θ_B (squares/triangles) with the linear path dashed. *Left to right*: no alignment, weight matching, activation matching, and weight matching with a REPAIR midpoint. Without alignment the path crosses a high red ridge (barrier 0.91); after alignment the endpoints lie in the same blue basin and the barrier drops to $\approx 1.1 \times 10^{-3}$.

shrinks activation variance, and corrects it by renormalising activation statistics. On the merging side, task arithmetic (Ilharco et al., 2023) defines task vectors $\tau = \theta_{\text{ft}} - \theta_0$ that can be added or subtracted, while TIES-merging (Yadav et al., 2023) reduces sign and magnitude interference between several task vectors. We use linear CKA (Kornblith et al., 2019) as a representational-similarity probe and 2D filter-normalised plots (Li et al., 2018) to visualise the landscape.

3. Method

Settings. We study three pairs chosen so that each isolates a different phenomenon: a three-hidden-layer **MLP-3** on **KMNIST** (Clanuwat et al., 2018) (no normalisation, pure permutation symmetry); a custom five-layer **Simple-ConvBN** with BatchNorm (Ioffe & Szegedy, 2015) on **SVHN** (Netzer et al., 2011) (tests BN handling and REPAIR); and **MobileNetV3-small** (Howard et al., 2019) fine-tuned on binary splits of **EuroSAT** (Helber et al., 2019) (a realistic backbone for task arithmetic).

Loss barrier. For two models we define the (relative) barrier

$$B(\theta_A, \theta_B) = \max_{\alpha \in [0,1]} \mathcal{L}(\theta(\alpha)) - [\alpha \mathcal{L}(\theta_A) + (1-\alpha) \mathcal{L}(\theta_B)], \quad (1)$$

the worst-case excess loss of the interpolated model over the endpoint average; $B \approx 0$ certifies linear connectivity. For n models we report the analogous barrier of the average $\bar{\theta} = \frac{1}{n} \sum_i \theta_i$ against the mean endpoint loss.

Alignment. To merge θ_B into the basin of θ_A we seek per-layer permutations $\{P_\ell\}$ that maximise neuron correspondence. *Weight matching* solves a LAP on the layer weights directly (Ainsworth et al., 2023); *activation matching* matches the cross-correlation of activations on a calibration batch; *Procrustes* aligns the (orthogonal, not permutation) subspaces as a baseline. Permutations are applied consistently to incoming and outgoing weights so the function of θ_B is unchanged.

REPAIR. Averaging two roughly independent activations halves their variance, $\text{Var}(\frac{h_A + h_B}{2}) \approx \frac{\sigma^2}{2}$, a *variance collapse* that compounds with depth. REPAIR (Jordan et al., 2023) rescales interpolated activations to restore the interpolated statistics; we also test the cheaper variant of only recalibrating BN running statistics, and a one-epoch fine-tune as an upper bound.

Multi-model and task arithmetic. We merge n models either by aligning all to a fixed *anchor* or *iteratively* folding each model into the running average. For editing we form task vectors $\tau_t = \theta_{\text{ft},t} - \theta_0$ and evaluate $\theta_0 + \lambda \sum_t \tau_t$ over a scaling sweep, comparing naive addition with TIES-merging (Yadav et al., 2023).

4. Experiments and results

Alignment removes the barrier. Table 1 reports barriers for two independently trained networks. For the normalisation-free MLP, weight and activation matching collapse the barrier by nearly three orders of magnitude ($0.91 \rightarrow 1.1 \times 10^{-3}$), confirming that the obstruction is almost entirely permutation symmetry. The orthogonal Procrustes baseline *increases* the barrier, since it leaves the permutation group and breaks the ReLU structure. For the BatchNorm ConvNet the effect is weaker but real ($2.03 \rightarrow 0.30$ with weight matching): convolutional channels and BN couple the symmetry and leave residual interference, which motivates REPAIR. Figure 1 visualises the same effect: the red ridge separating the two minima disappears after alignment.

Table 1. Linear interpolation barrier B between two independently trained models (lower is better). WM/AM = weight/activation matching.

	Naive	WM	AM	Procrustes
MLP-3 / KMNIST	0.911	0.001	0.001	1.416
ConvBN / SVHN	2.027	0.296	1.125	2.158

Variance collapse and REPAIR. The per-layer activation-variance ratio along the interpolation (Figure 4, appendix) is symmetric, deepest at $\alpha = 0.5$, and the fully-connected layers fall close to the theoretical $1/\sqrt{2} \approx 0.707$ floor, confirming the variance-halving mechanism. Correcting it pays off: on ConvBN/SVHN, BatchNorm recalibration alone lowers the merged loss from 0.49 to 0.42 at essentially zero cost, while a single fine-tuning epoch reaches 0.94 accuracy. Full REPAIR helps the MLP but, on this small ConvNet, over-rescaling slightly hurt the loss, so the cheap BN recalibration was the best practical choice.

Merging many models. Figure 2 scales merging to $n \in \{3, 5, 8\}$ MLPs over eight seeds. Naive averaging degrades fast: merged accuracy falls from 0.51 ($n = 3$) to 0.10 ($n = 8$, i.e. chance) as the barrier grows. Anchor alignment keeps accuracy at ≈ 0.88 , and *iterative* alignment is best, holding ≈ 0.89 accuracy with a barrier that stays slightly *negative* even at eight models—the merged network is as good as the endpoints. The iterative merge is also order-invariant: across all $3!$ orderings of three models the barrier is -0.070 ± 0.009 . TIES and SLERP, designed for fine-tuned (not independently trained) models, fail here, with TIES diverging—underlining that alignment, not interference resolution, is the operative factor for from-scratch models.

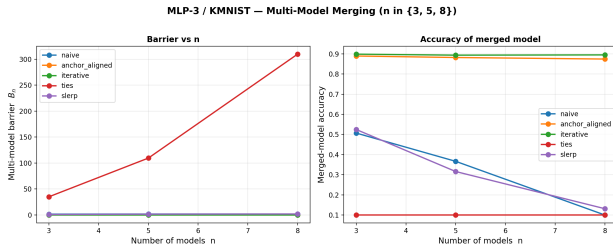


Figure 2. Multi-model merging (MLP-3 / KMNIST, $n \in \{3, 5, 8\}$). Left: barrier vs. n . Right: merged accuracy. Iterative and anchor alignment stay flat; naive and SLERP collapse; TIES diverges.

Task arithmetic. On MobileNetV3-small/EuroSAT the two binary tasks reach 99.2% and 98.7% individually. Naively summing both task vectors ($\lambda = 1$) collapses joint accuracy to chance (51%) and the loss diverges for $\lambda \geq 0.8$ (Figure 3). Scaling down the task vectors restores cooperation: the best scaling $\lambda = 0.6$ gives 86.9% joint accuracy, and TIES-merging reaches a comparable 83.8% by sparsifying and sign-aligning the vectors. Thus the same interference that alignment fixes for hidden-unit permutations re-appears as magnitude interference between task vectors, and is again controlled by a simple scaling.

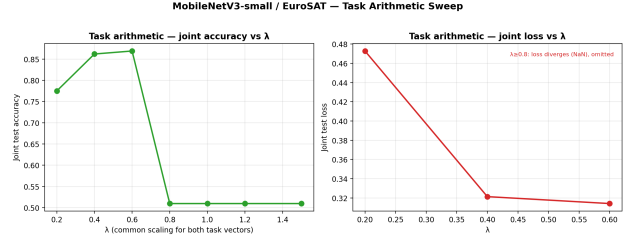


Figure 3. Task arithmetic on MobileNetV3-small / EuroSAT. Joint accuracy (left) peaks at $\lambda = 0.6$; the joint loss (right) diverges for $\lambda \geq 0.8$.

5. Discussion and conclusions

Across three architectures we find a consistent picture. The linear barrier between independently trained networks is largely a permutation artefact: a per-layer assignment removes almost all of it for a normalisation-free MLP, and a substantial part for a BatchNorm ConvNet, where the residue is explained by activation variance collapse and cured cheaply by BN recalibration. Alignment scales gracefully—iterative merging keeps eight models in a single basin—whereas methods built for fine-tuned models (TIES, SLERP) need a shared initialisation to work, which our spawning control (children forked from a common parent show near-zero barrier even *without* alignment) confirms. Finally, task arithmetic shows the same lesson in weight space at the level of task vectors: scaling, or interference resolution via TIES, turns a chance-level merge into one within a few points of single-task accuracy. The main limitation is scale—our backbones are small and our merges involve few models—so extending the iterative-alignment result to larger networks and to fine-tuned foundation models is the natural next step.

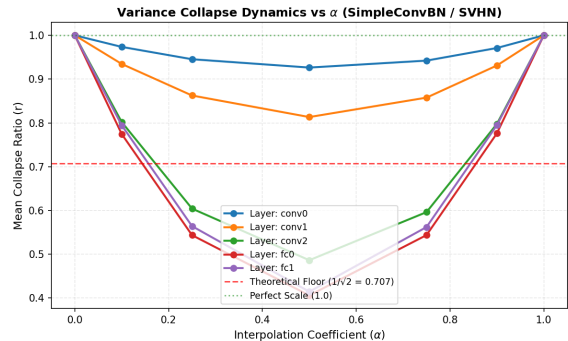


Figure 4. Per-layer variance-collapse ratio r vs. interpolation coefficient α (SimpleConvBN / SVHN). Deeper layers collapse towards the $1/\sqrt{2}$ floor at the midpoint, the effect REPAIR corrects.

6. Statement on the use of AI

At the onset of this project, our primary goal was to deepen our understanding of the subject matter, as our initial knowledge was still developing. We began by thoroughly researching the problem space, consistently creating foundational drafts for both our codebase and this report. To accelerate our learning process and refine these drafts, we extensively utilized AI-based tooling (specifically, an LLM-powered coding assistant).

Throughout the project, AI served as a collaborative tool to: (i) help implement and debug our core code drafts, including the loss-barrier and CKA metrics, weight-/activation-matching, Procrustes alignment routines, REPAIR variants, and merging loops; (ii) prepare experiment scripts and the plotting code that produced the figures in this report; (iii) assist with data loading and preprocessing for KMNIST, SVHN, and EuroSAT; (iv) explore relevant literature to locate cited references; and (v) edit and polish our text drafts for clarity and page-limit constraints. Additionally, as non-native English speakers, we found the AI invaluable for correcting grammar and naturally refining the language of our initial drafts across both the codebase and the final report.

We take full responsibility for the submitted material. By actively guiding the project through every phase, we have ensured a deep, comprehensive understanding of our methodology and results. While AI significantly boosted our drafting and development process, the experimental design, the choice of architectures and datasets, the selection of hypotheses to test, and the interpretation of all results remain strictly our own. Furthermore, it is important to note that while current AI excels at writing code and facilitating learning, it lacks the capacity to autonomously execute experiments, synthesize empirical outcomes, or grasp the deeper "why" behind the results. Consequently, the extensive hours we dedicated to hands-on trial and error, rigorous testing, and critical analysis remain fundamentally irreplaceable. We have independently verified every reported number against the raw experiment outputs, re-ran key experiments across multiple seeds, and rigorously checked that the final code accurately produces the stated results.

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Appendix

Training details. MLP-3: hidden width 256, ReLU, no normalisation, trained with SGD on KMNIST. Simple-ConvBN: three 3×3 conv blocks (32/64/128 channels) with BatchNorm and max-pooling, then a 256-unit BN-FC head, trained on SVHN. MobileNetV3-small: ImageNet-initialised backbone fine-tuned on binary EuroSAT splits (T1: vegetation vs. built-up; T2: water vs. land). Barriers use 11 interpolation points; activation/variance statistics use 10 calibration batches.

REPAIR variants (ConvBN / SVHN). No repair: loss 0.493, acc 87.0%. BN recalibration: loss 0.420, acc 87.2%. Full REPAIR: loss 1.324, acc 86.2%. One-epoch fine-tune (upper bound): loss 0.211, acc 94.2%.

Spawning control. Two children forked from a shared parent after k common epochs and then trained independently show barriers of $\sim 10^{-3}$ for all $k \geq 1$ (vs. 0.088 for $k = 0$), even without alignment, confirming that a shared early trajectory is sufficient for linear connectivity.